



Debug the Pair

One script is wrong — find it and fix it

Name: _____

Date: _____

★ I can find which of two scripts is wrong and fix it so the sprites meet.

The goal — and the bug

Bit (green) and Pixel (purple) are SUPPOSED to MEET on number 3 when the green flag ► starts both. But when you run them, they do NOT meet — one script has a bug! Read both scripts, find the wrong one, and fix it so they meet on 3.

The goal: meet on 3



Bit's script

► start at 0

forward 3

Pixel's script

► start at 5

back 1

1 Run it — what goes wrong?

Run both scripts. Bit: ____ Pixel: ____ . Do they meet on 3? Circle YES or NO.

2 Find the bug

Whose script is wrong — Bit's or Pixel's? (Hint: one sprite already lands on 3.) Write whose script has the bug.

3 Fix it

Write the fixed move for the wrong script so BOTH sprites meet on 3 when the green flag starts them.

Fixed move: _____

Big idea

Debugging two scripts means checking each one against the goal. Bit already lands on 3, so the bug is in Pixel's script: changing 'back 1' to 'back 2' moves Pixel to 3, and now the pair meets.



Debug the Pair

Quick facilitation notes & answer key

What this worksheet teaches

The challenge sheet: students DEBUG a pair of move sets. The two characters are meant to meet on 3, but Pixel lands on 4. Students check each set against the goal, decide which one is wrong, and fix it. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: put two counters on a number line 0 to 5 — green for Bit, purple for Pixel — and move BOTH at the same time so children see the two sets of moves run side by side.

How to lead it (about 20–25 minutes)

1. Show them (you do). Run both; they do not meet on 3 — one set has a bug.
2. Do one together. Exercise 1 — Bit lands on 3, Pixel on 4 (they miss). Exercise 2 — Bit already reaches 3, so Pixel's set is the buggy one.
3. Let them try. Students write the fixed move so both meet on 3 (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 0, forward 3 → 3. Pixel: 5, back 1 → 4. They do not meet on 3 (Pixel is on 4).

Exercise 2 — Pixel's set is the wrong one; Bit already lands on the target 3, so the bug must be in Pixel's.

Exercise 3 — fix Pixel to back 2: 5 → 3. Now both land on 3 → they meet. (Accept any Pixel move that lands on 3, e.g. back 1, back 1.)

If students get stuck — and ways to adjust

Watch for: changing both sets. Only the buggy one (Pixel's) needs fixing.

Make it easier: check which character already reaches the target, then fix the other.

Make it harder: ask for a two-move Pixel fix that also lands on 3.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, change one, and explain what happens when both run together.

You'll know it worked when

The child finds Pixel's set as the buggy one and fixes it so both meet on 3.